Mason Bane

817-487-5148 | mbane0525@gmail.com | linkedin.com/in/mason-bane/ | mbane04.github.io

Summary: Computer Science and Engineering student with hands-on experience in Linux systems administration, hardware diagnostics, and infrastructure automation. Combines strong troubleshooting skills in hardware and Linux environments with scripting proficiency in Python and Bash. Proven ability to maintain high-availability systems, document processes, and collaborate in technical operations roles. Seeking to apply operational rigor and technical versatility to large-scale data center environments.

Education

Tarleton State University — Stephenville, TX
Bachelor of Science in Computer Science, Concentration in Computer Engineering — *Aug. 2022 – May 2026 (Expected)*
Minor in Mathematics
GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Hill College — Hillsboro, TX
Associate of Arts in Liberal Arts — *Aug. 2019 – Sept. 2023*

Experience

Undergraduate Research Assistant - Systems & Performance — May 2024 – Present
Tarleton State University — *Stephenville, TX*

- Developed and optimized high-performance computing simulations in C/C++, implementing parallel processing algorithms to maximize computational efficiency on Linux clusters.
- Engineered robust, maintainable software through systematic testing and debugging protocols, ensuring reliability for long-running computational workloads.
- Created comprehensive documentation for software deployment, configuration, and troubleshooting, facilitating collaboration and knowledge transfer across research teams.

Undergraduate Technology Specialist - HPC Lab Manager — Aug. 2024 – Present
Tarleton State University — *Stephenville, TX*

- Managed 15+ Linux systems in production-like research environment, ensuring 99%+ uptime through proactive maintenance, hardware diagnostics, and rapid incident response
- Diagnosed and resolved complex hardware/software issues including memory failures, storage degradation, and OS configuration problems, minimizing system downtime and maintaining optimal performance
- Developed and maintained comprehensive documentation for system configurations, troubleshooting procedures, and lab protocols, enabling knowledge sharing and process standardization

Projects

Secure IoT Sensor Node with Hardware TrustZone | *C, STM32CubeIDE/HAL, I2C/SPI* — Dec. 2025 – Present

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.
- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

Real-Time Cardiac Simulation Engine | *NIH Grant #1R15HL179671-01* | *C/C++, Linux* — Aug. 2024 – Present

- Developed real-time 20,000+ node cardiac model for Linux systems, implementing performance optimizations through parallel algorithms and creating accessible, documented interfaces for end users
- Automated the simulation toolchain with Bash scripts for compilation, testing, and data validation across multiple Linux workstations, ensuring reproducibility and reducing manual setup time
- Developed and maintained comprehensive documentation for system architecture, build procedures, and troubleshooting protocols to support collaboration and future research initiatives

Technical Skills

Linux & Systems Administration: Linux (Administration, Troubleshooting), Git, CMake/Make
Hardware & Server Support: Server Hardware Concepts, Hardware Diagnostics, Oscilloscope, Multimeter, GDB
Automation & Scripting: Python, Bash, JavaScript
Networking Concepts: DNS, HTTP, DHCP, RAID (I^2C, SPI, UART)
Programming & Tools: C, C++, ARM Assembly, Java, HTML/CSS